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NONLINEAR EFFECTS OF AI ADOPTION AND DEVELOPMENT ON UNEMPLOYMENT: THRESHOLD MODELS IN A GLOBAL PANEL

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ABSTRACT

We examine the relationship between AI innovation and aggregate unemployment using a dynamic panel Fixed Effects (OLS FE) framework, supplemented by a Anderson-Hsiao Instrumental Variable (AH-IV) estimation as a robustness check. Using AI patent filings from 36 countries across 1991-2020, we decompose AI adoption into an initial "AI onset jump" and subsequent innovation intensity. Our baseline results find no statistically significant contemporaneous effect of AI adoption on unemployment, suggesting that labour market responses are not universally immediate. However, our endogenous threshold model identifies a regime split in effects at USD 24,564 GDP per capita, with evidence of heterogeneous regime-specific effects. In lower-income countries, AI innovation is associated with a statistically significant cumulative reduction in unemployment, suggesting that the labour-absorbing reinstatement effect dominates in these economies. Conversely, the cumulative effect in high-income economies is only marginally significant ($p < 0.1$), suggesting intensive task reallocation rather than net headcount expansion. These findings suggest that AI's labour market impacts are development-dependent, driven by a mix of displacement and reinstatement effects.

JEL Classification: C23, C24, E24, J23, O33

Keywords: Artificial Intelligence, Unemployment, Threshold Regression, Innovation, Patents

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1. INTRODUCTION

In recent years, artificial intelligence (AI)'s impacts on labour markets, especially unemployment, have become an increasingly contentious topic. Although AI often raises productivity, it can also automate certain tasks, and the balance between these two forces is not always clear (Acemoglu and Restrepo, 2019). Because of this, countries may not experience the same employment effects from AI, and the timing of these effects may also differ.

New technologies like AI rarely influence the economy immediately. Firms usually need time to reorganise production, train workers, and build complementary systems before they begin using AI effectively (Kydland & Prescott, 1982; Elhanan Helpman, 2010). At the same time, economies differ greatly in their structure and income levels, and these differences may shape whether AI ends up creating or replacing jobs (Nguyen and Vo, 2022; Wang et al., 2024; Bouattour, 2023; Çetin, 2025).

Given these uncertainties, it is important to examine how AI innovation unfolds over time and whether its effects vary across countries. This project uses AI-related patent filings as an indicator of domestic AI development, allowing us to track when a country first becomes active in AI innovation and how its activity evolves. Patents offer a way to observe the early stages of technological capability (OECD, 2009), which makes them useful for studying both the onset and intensity of AI development.

To guide the analysis, this study focuses on two central research questions:

- Research Question 1 (Onset / Early Adoption Effect):

Does AI adoption, measured using patent activity, affect unemployment after a country begins developing AI?
This helps identify whether the initial shift into AI innovation leads to noticeable changes in labour-market outcomes.

- Research Question 2 (Threshold / Nonlinearity Among AI-Active Countries):

Among countries that have already begun innovating in AI, is there a threshold level of economic development that changes how AI intensity influences unemployment?
Here, development level (captured by lagged log GDP per capita) is examined as a possible dividing line that separates countries into different adjustment "regimes".

Furthermore, by analysing the temporal distribution of AI's impact, we contribute to the General Purpose Technology (GPT) literature by evaluating whether the intangible, software-driven nature of AI results in a more compressed "time-to-build" lag compared to historical physical technologies.

Overall, this project aims to provide a clearer understanding of how AI innovation relates to unemployment across a diverse set of countries. By considering both delayed effects and differences based on development levels, the study offers a more realistic view of how economies might respond to the ongoing rise of AI technologies.

2. LITERATURE REVIEW

2.1 Theoretical Basis

The relationship between AI and unemployment is well grounded in modern theories of technological change, particularly task based theories of automation and labor demand (e.g. Acemoglu and Restrepo, 2019). In contemporary labor economics, AI is modeled as a task changing technology that alters which tasks are carried out by workers versus machines, similar to how robots changed employment structures (Acemoglu and Restrepo, 2016; Acemoglu and Restrepo, 2019). This task reallocation provides a direct link between AI adoption and employment outcomes. When AI expands the set of automatable tasks, labor demand may fall in affected activities. Even if total output rises, fewer workers may be required in the automated portion of the task bundle, making unemployment, and not only productivity growth, a key outcome to examine.

2.1.1 Task-Based Mechanisms and Theoretical Ambiguity

Acemoglu and Restrepo (2019) formalises three key forces that operate simultaneously:

- **Displacement effect**: automation substitutes for workers in tasks they previously performed, reducing labor demand.
- **Productivity effect**: automation lowers production costs, expands output, and can raise labor demand in complementary or non-automated tasks.
- **Reinstatement**: innovation can generate new tasks in which labor has a comparative advantage, partially restoring labor demand over time.

These forces point in different directions; hence, the framework does not predict a clear universal effect of AI on unemployment. Which mechanism dominates is likely to differ across countries depending on structural characteristics such as sector composition, human capital, skill distribution, and institutions that shape adjustment. Consistent with this view, the International Labour Organization (2024) suggests that advanced economies face both higher exposure to task automation and greater potential for task augmentation. This supports the relevance of cross country analysis in which different development regimes may lead to different dominant mechanisms.

Recent empirical evidence reinforces this conditional view. Çetin and Kutlu (2025) uses System-GMM estimation on 29 countries (2017–2021) and reports a positive but fragile relationship between AI and employment: the effect becomes statistically insignificant when labor productivity interactions are introduced. This underscores that productivity gains, one of the channels emphasized in task-based frameworks, may offset or reverse simple correlations, reinforcing the need to model AI's labor-market impact as conditional rather than universal.

A complementary perspective frames AI as an “advance in prediction” rather than simply automation. Agrawal, Gans, and Goldfarb (2019) argue that as prediction becomes cheaper and more accurate, some prediction intensive tasks are substituted away while other activities become more valuable. This is consistent with the task based view because substitution and complementarity can occur at the same time. In other words, AI plausibly affects employment, but the direction and magnitude depend on how production and organisations reorganise around the technology.

Collectively, AI can affect unemployment through multiple competing channels, making the net effect theoretically ambiguous and likely to vary across countries.

2.1.2 AI as a General Purpose Technology and Delayed Labour Market Adjustment

An important extension for the present study is the treatment of AI as a General Purpose Technology (GPT). The GPT literature implies that the effects of transformative technologies on aggregate outcomes are often delayed rather than immediate. Bresnahan and Trajtenberg (1995) argue that GPTs require complementary investments in organisational design, human capital, and supporting systems before productivity gains can be broadly realised.

This delayed transmission from inventive activity to economic outcomes is well documented. Griliches (1990) shows that patent-based indicators correlate with productivity only after innovations have diffused and been commercially implemented. More broadly, "time-to-build" frictions mean that the integration of new technologies and the associated reallocation of labor and capital unfold over several periods rather than instantaneously (Kydland & Prescott, 1982; Elhanan Helpman, 2010). This timing logic motivates the inclusion of lagged AI patent intensity in our baseline specification. More significantly, in our sample period, the proliferation of cloud computing, open-source machine learning frameworks (e.g., TensorFlow) and API-based deployment drastically lowered the technological barriers to entry (Delipetrev et al., 2020), which we believe would have reduced the initial “time-to-build” lag for AI. Additionally, from an econometrics perspective, including many adjacent annual lags of a persistent patent series tends to inflate multicollinearity and reduce finite-sample power, producing unstable and difficult-to-interpret coefficients. Therefore we propose a short lag structure of 1-2 years, which captures the near-term diffusion window suggested by the literature while preserving the precision and interpretability of the estimated employment responses.

The empirical difficulty of capturing these delayed adjustments is also consistent with OECD (2023), which notes in Employment Outlook 2023 that aggregate employment effects of AI are so far hard to find. This may reflect short observation windows, gradual adoption, and the possibility that displacement and productivity effects offset each other in the short run. From an econometric perspective, evaluating "time-to-build" frictions also necessitates the use of distributed lags, which consumes degrees of freedom and motivates an extended, long-horizon cross-country panel. Therefore, our 30-year, cross-country panel approach is a necessary addition to literature to evaluate the conditions under which AI effects on the labor market emerge and whether they differ across stages of economic development.

Overall, task based mechanisms and GPT style adjustment dynamics suggest that AI related effects on unemployment may be nonlinear, delayed, and dependent on the level of economic development. This supports the use of both distributed lag and threshold approaches in this study.

2.2 Patent-Based Measures of AI Innovation

Patent data are widely used as indicators of technological innovation because they capture outputs of inventive activity, rather than only inputs such as R&D expenditure. The OECD's *Patent Statistics Manual* identifies patents as one of the most frequently used indicators of technological output (OECD, 2009). For AI specifically, patent-based measurement is supported by institutional practice. World Intellectual Property Organization's *Technology Trends 2019: Artificial Intelligence* analyses AI innovation using patent application data to map global trends, demonstrating that "AI patents" are an accepted empirical measure of AI innovation (WIPO, 2019).

In cross‑country settings, especially those including middle‑income economies, patent data typically display many zeros in early periods. Continuous patent counts are therefore highly skewed and dominated by frontier economies. This makes the extensive margin, the first observable AI patent, informative. We treat this as an AI‑innovation onset indicator, signalling the first domestically documented, codified AI invention. This does not measure broad diffusion but rather the initial emergence of AI inventive capability, making it suitable for threshold and nonlinear analysis.

A key concern is whether patent-based indicators contain economically meaningful signals. Macro evidence shows that AI patent data can be used to extract real, macro-relevant information (Evgenidis & Fasianos, 2025). The paper constructs a quarterly measure of AI innovation, then identifies exogenous AI "news shocks", findings that raise expectations about future AI progress. They find that positive AI news shocks generate expansionary macro responses (e.g., higher output, investment, wages, and hours) and reduce unemployment. This shows that AI patent-based indicators can be incorporated into macro frameworks to identify AI-related technological expectations with measurable labor-market implications.

2.3 Related Works in Thresholds

Recent empirical work suggests that the link between AI and unemployment is unlikely to be uniform across contexts. Several cross-country panel studies have documented threshold effects and regime-dependent relationships, providing crucial context for our analytical approach.

Nguyen and Vo (2022) employ a Panel Smooth Transition Regression (PSTR) model across 40 developed and emerging economies (2000–2019) to examine how AI affects unemployment. They find that inflation acts as a threshold variable: AI's employment impact differs systematically depending on whether inflation exceeds certain critical levels. This suggests that macroeconomic stability conditions influence how AI technologies reshape labor demand, consistent with task-based theories where displacement and productivity effects may be amplified or dampened by the broader economic environment.

Another study by Wang et al. (2024) analyses Chinese provincial data using panel threshold regression, revealing that AI's influence on employment structure intensifies only after industrial development surpasses certain maturity thresholds. This suggests that the stage of economic transformation, captured by industrial structure optimization, mediates how AI adoption translates into labor-market adjustments. Similarly, Bouattour (2023) applies nonlinear threshold methods to technology imports across developed and developing countries, confirming that development level itself constitutes a meaningful regime boundary: the employment effects of imported technology differ sharply between high- and middle-income settings.

Overall, these studies establish that AI's employment effects are context-dependent, moderated by macroeconomic conditions, structural factors, and dynamic adjustments. Building on these related works, we use GDP per capita, acting as a proxy for a nation's aggregate development level and structural maturity, as the endogenous threshold variable.

By employing Hansen's (1999) threshold framework, we provide a comprehensive assessment of where and when AI-related technological activity exhibits regime-specific associations with unemployment.

3. METHODOLOGY

3.1 Data Sources and Variables

All macroeconomic variables used in this study are drawn from the World Bank's World Development Indicators (WDI) to ensure cross-country comparability and consistent definitions over time. In particular, we derive the unemployment rate as our primary labor-market outcome, and compile additional covariates that capture macro conditions and technology-linked channels commonly highlighted in the literature: patent activity (as our main AI-related innovation proxy), foreign direct investment (FDI), inflation, population, government expenses, GDP per capita.

Our empirical sample is constructed as a balanced panel spanning 1991–2020, selected to maximise internal comparability and to support dynamic and nonlinear models. The panel includes 36 countries: Australia, Austria, Bulgaria, Canada, Switzerland, Chile, Costa Rica, Cyprus, Germany, Denmark, Finland, France, United Kingdom, Guatemala, Croatia, Hungary, Ireland, Iceland, Israel, Italy, Jamaica, Jordan, Korea, Sri Lanka, Malta, Netherlands, Norway, Peru, Portugal, Romania, Singapore, Sweden, Thailand, Uruguay, United States, South Africa. This country set is deliberately heterogeneous in development level and economic structure, allowing us to test the central hypothesis emerging from the literature review: that the relationship between AI-related technological activity and unemployment is unlikely to be uniform, and may differ systematically across development regimes because displacement, productivity, and reallocation forces are mediated by skills, sector composition, and institutional capacity.

The following variables were constructed from the raw WDI and WIPO data:

Table 1. Description of Dependent Variables

Variables	Source(s)	Definition
Unemployment Rate	*World Bank*	*Modeled unemployment, ILO estimate (% of total labor force). Log-transformed to normalise the distribution.*

Table 2. Description of Independent Variables

Variables	Source(s)	Definition
AI Innovation Proxy	*World Intellectual Property Organization WIPO*	*Evaluated as log(patents + 1). Log-transformed WIPO AI-related Patent Count to address right-skewness. Adjustment of +1 to handle zero-patent observations common in some economies during early sample years.*
Development Threshold	*World Bank*	*GDP per capita (constant 2017 PPP $). Serves as the primary endogenous threshold variable indicating the level of economic development. Log-transformed to normalise the distribution.*
Inflation	*World Bank*	*Annual percentage change in consumer price index. Included to control for macroeconomic stability and nominal shocks.*
Government Expenditure	*World Bank*	*General government final consumption expenditure (% of GDP). Included to control for varying fiscal policy environments and state intervention.*
Foreign Direct Investment	*World Bank*	*Net FDI inflows (% of GDP). Included to control for international technology transfer, capital flows and global integration.*
Population Scale	*World Bank*	*Measured as* $log(population)$*. Included to control for absolute demographic scale and labor market size.*

3.2 Pre-Estimation Diagnostics and Transformations

Prior to specifying the dynamic panel and threshold models, the dataset must be evaluated for standard panel assumption violations. Specifically, we tested for cross-sectional dependence and panel unit roots for evidence of non-stationarity and shared global shocks, which are common issues in panel macroeconomic data. These diagnostics guided our decision to implement first-differencing to enforce stationarity and eliminate any time-invariant country fixed effects.

3.2.1 Cross-Sectional Dependence Testing

Macroeconomic panel data across open economies frequently exhibit cross-sectional dependence due to shared global shocks, financial integration and trade linkages. This is especially relevant for our dataset as it spans across multiple global financial shocks between 1991 and 2020. Therefore, we first tested the residuals of a baseline fixed-effects regression using Pesaran's (2004) Cross-Sectional Dependence (CD) test. Detecting this potential significant cross-panel correlation was critical to avoid biased standard error estimates. Based on our results in Table 3, we rejected cross-sectional independence for all variables, suggesting that unemployment shocks are correlated across countries due to global business cycles, trade linkages and financial integration. Therefore, in our subsequent estimation, we at minimum used country-clustered standard errors (heteroskedasticity robust) and year dummies (common time shocks).

Table 3. Pesaran CD Results

Variable	N	Average T	Average Rho	CD Statistic	P Value
log_unemp	36	30.000	0.0704	9.682	0.000
log1p_patents	36	29.712	0.498	53.307	0.000
log_pop	36	30.000	0.535	73.491	0.000
log_gdppc	36	30.000	0.886	121.871	0.000
Inflation	36	30.000	0.478	65.689	0.000
Govt Expense	36	30.000	0.133	18.225	0.000
FDI	36	30.000	0.234	32.202	0.000

3.2.2 Panel Unit Root Testing

To limit the risk of spurious regressions, the time-series properties of the variables are assessed using the Cross-Sectionally Augmented IPS (CIPS) test, supported by a fallback Fisher-ADF test, to explicitly account for cross-sectional dependence identified in Step 1. Classifications of each variable are as below:

Table 4. CIPS Results

Variable	**Order of Integration**
Unemployment Rate	I(1)
AI Innovation Proxy	I(0)
Labor Productivity/GDPpc	I(1)
Inflation	I(0)
Government Expenditure	I(1)
Foreign Direct Investment	I(0)
Population Scale	I(1)

3.2.3 First-Differencing (FD) Transformation

Guided by the unit root diagnostics, the baseline model is transformed using First Differences (FD). All variables were differenced once. As seen in Table 5, these variables were then tested using the panel unit test to ensure all regressors in the final estimation matrix are stationary.

Table 5. Level and FD Unit Root Check Results

Variable	P-Value
d_log_unemp (level)	<0.001***
log1p_patents (level)	0.009***
log_gdppc (level)	0.263
log_pop (level)	0.382
Inflation (level)	<0.001***
Govt Expense (level)	0.051*
FDI (level)	<0.001***
d_log_unemp (FD)	<0.001***
d_log1p_patents (FD)	<0.001***
d_log_gdppc (FD)	<0.001***
d_log_pop (FD)	<0.001***
d_Inflation (FD)	<0.001***
d_Govt Expense (FD)	<0.001***
d_FDI (FD)	<0.001***

(***) p<0.01, (**) p<0.05, (*) p<0.10

This transformation enforces stationarity across the panel and mathematically eliminates unobserved, time-invariant country fixed effects, such as geographical advantages, that could confound the relationship between AI onset adoption and employment dynamics.

3.3 Empirical Specifications and Model Variables

The labour market is characterised by high persistence, so current unemployment is heavily dependent on past levels due to hysteresis (Blanchard and Summers, 1986). Therefore, the baseline labor market model specification needs to be dynamic. However, in a standard dynamic panel, the lagged dependent variable correlates with the unobserved country-specific fixed effect (α_i). After our FD transformation, we successfully eliminated α_i, but this also mathematically induced a new correlation between the differenced lagged dependent variable ($\Delta y_{i,t-1}$) and the differenced error term ($\Delta\epsilon_{it}$), which may generate Nickell Bias (Nickell, 1981). However, since Nickell bias is most significant in panel data for which N>>T and the order of the bias is of order $\frac{1}{T}$, our post-onset panel with $T \approx 25$ reduced the estimated bias to approximately 4%. Consequently, we selected the Ordinary Least Squares (OLS) Fixed Effects (FE) estimator as our primary specification to preserve statistical efficiency and precision in our post-onset $N = 32$ sample. To ensure our results are not an artifact of residual dynamic correlation, we employed the Anderson-Hsiao Instrumental Variable (AH-IV) estimator (Anderson and Hsiao, 1981) as a robustness check.

3.3.1 Onset vs. Intensity Specification (RQ1)

To evaluate whether AI adoption affects unemployment after a country's initial AI onset, the model needed to distinguish between the structural shift of initial adoption and the ongoing intensity of innovation. First, we defined a country's annual AI patent activity indicator, A_{it}, which equals 1 if the country produces any AI patents in year t, and 0 otherwise. From this, we constructed the cumulative adoption state, S_{it}, which equals 1 if a country has ever produced an AI patent up to year t, and 0 otherwise. From this base state, the AI patent proxy is decomposed into two distinct variables to avoid attributing the "first adoption year" discontinuity to intensity changes:

- Onset Jump (d_S_{it}): A binary indicator capturing the exact year a country transitions from zero prior AI patents to producing its first AI patent(s). Because the empirical panel only spans the period 1991-2020, countries that already exhibited AI patenting activity in or prior to 1991, as well as countries that never produced any AI patents in and prior to 2020, do not have an observable onset jump within the sample timeframe (d_S_{it} evaluates to 0 for all observable first-differences). Consequently, the baseline onset effect was identified exclusively from the 32 countries that transitioned into AI adoption during the sample period.
- Post-Onset Intensity ($AI\_post\_intensity_{it}$): An interaction term multiplying the first-differenced AI innovation proxy $\Delta log(patents + 1)$ by the lagged adoption state $S_{i,t-1}$, i.e. $AI\_post\_intensity_{it} = S_{i,t-1} \times \Delta log(patents + 1)$.

The dynamic OLS FE specification for RQ1 is estimated as:

$$\Delta y_{it} = \alpha\Delta y_{i,t-1} + \beta_1 d\_S_{it} + \beta_2 AI\_post\_intensity_{it} + \phi\Delta X_{it} + \Delta\tau_t + \Delta\epsilon_{it}$$

Where:

- Δy_{it} : the first-differenced log unemployment rate for country i in year t.
- d_S_{it}: the binary indicator for Onset Jump.
- $AI\_post\_intensity_{it}$: the interaction term for Post-Onset Intensity.
- ΔX_{it} : the vector of first-differenced macroeconomic controls (inflation, government expenditure, FDI and population scale).
- $\Delta\tau_t$: year fixed effects to absorb common global macroeconomic shocks.

In this model, the coefficient β_2 tests RQ1 by isolating the marginal effect of accumulating AI patents strictly within post-onset country-years.

3.3.2 Threshold Identification and Nonlinearity (RQ2)

To investigate whether there is a nonlinear threshold break in the effect of AI intensity based on economic development, the sample is restricted exclusively to post-onset observations ($S_{i,t-1} = 1$). Since the displacement and reinstatement mechanisms of AI are highly sensitive to structural contexts (Acemoglu and Restrepo, 2019), forcing a uniform β_2 parameter across all countries may obscure the true macroeconomic dynamics.

Testing only contemporaneous effects of AI risks Omitted Variable Bias (OVB) due to the macroeconomic frictions in technological diffusion. Therefore, we introduced a distributed lag model incorporating both contemporaneous (ΔAI_{it}) and one-year lagged ($\Delta AI_{i,t-1}$) patent intensity based on two fundamental justifications:

- Economic Frictions ("Time-to-Build"): This "Time-to-Build" lag (Kydland & Prescott, 1982; Elhanan Helpman, 2010) implies that current employment fluctuations are simultaneously driven by both current and delayed commercialisation shocks.
- Econometric OVB Conditions: Lagged AI patenting is highly autocorrelated with current patenting and independently affects unemployment. Forcing multi-year labour adjustments into a single contemporaneous observation winder can artificially conflate these distinct shocks in the model. This omission can drive the estimated coefficients toward zero, masking true structural employment shifts as noise.

A potential concern with including the contemporaneous AI intensity is reverse causality: current unemployment could influence a country's patenting activity. However, patents reflect inventive output that typically requires several years of R&D, development and legal processing before filing (Popp et al., 2004) mitigating simultaneity bias between patenting activity and current unemployment.

Rather than imposing an arbitrary, subjective split between high- and low-income countries, we adapted the Hansen (1999) Threshold Grid Search methodology to find the exact threshold break endogenously. The primary nonlinear distributed lag model specification is:

$$\Delta y_{it} = \alpha \Delta y_{i,t-1} + \left(\sum_{k=0}^{2} \theta_{k,L} \Delta AI_{i,t-k}\right) \cdot I(\pi_{it} \leq \gamma) + \left(\sum_{k=0}^{2} \theta_{k,H} \Delta AI_{i,t-k}\right) \cdot I(\pi_{it} > \gamma) + \phi \Delta X_{it} + \Delta \tau_t + \Delta \epsilon_{it}$$

Where:

- $\theta_{k,L}$: the coefficients for AI intensity in the Low (L) development regime at lag k, where $k = 0$ is contemporaneous, $k = 1$ is the 1-year lag, and $k = 2$ is the 2-year lag.
- $\theta_{k,H}$: the coefficients for AI intensity in the High (H) development regime at lag k, where $k = 0$ is contemporaneous, $k = 1$ is the 1-year lag, and $k = 2$ is the 2-year lag.
- $\Delta AI_{i,t-k}$: the first-differenced AI patent intensity at lag index k.
- $I(\cdot)$: the indicator function, which equals 1 if the condition inside the parentheses is true, and 0 otherwise.
- π_{it} : the threshold variable (lagged log GDP per capita $L1\_log\_gdppc$).
- γ : the endogenous threshold parameter representing the income boundary between regimes.

This model tests RQ2 by allowing the slope coefficient for the AI Innovation Proxy to vary depending on the established macroeconomic boundary, yielding four distinct estimates: $\theta_{0,L}$ and $\theta_{1,L}$ representing the contemporaneous and lagged shocks within the low-development regime; and $\theta_{0,H}$ and $\theta_{1,H}$ representing the corresponding shocks within the high-development regimes. This model evaluates whether the structural relationship between technological innovation and labour market outcomes undergoes a structural break both immediately and over the 1-to-2-year adjustment period.

3.4 Estimation Strategy

3.4.1 Dynamic Panel Estimation and Instrument Validity

The baseline estimation used an OLS FE estimator. Prior to final estimation, a robust Woolridge test for panel data confirmed the presence of first-order serial correlation (AR(1)) within countries ($p < 0.001$). Therefore, all standard errors were clustered at the country level to ensure valid inference.

3.4.2 Endogenous Threshold Identification

To test for structural heterogeneity (RQ2), we adapted the Hansen (1999) Threshold Grid Search methodologies for instrument variables. Rather than imposing a subjective boundary between development levels (such as based on World Bank's income group classification), our algorithm executed an exhaustive grid search over all unique values of the threshold variable (lagged log GDP per capita π) in the post-onset sample.

To preserve sufficient degrees of freedom and avoid identifying small, unrepresentative regimes, the search grid was trimmed at the top and bottom 10% of the threshold variable's distribution. For each candidate threshold γ in the trimmed grid, the OLS FE model was estimated. The optimal structural break $\hat{\gamma}$ was identified at the specific macroeconomic boundary that strictly minimised the concentrated sum of squared residuals (SSR).

3.4.3 Regime-Specific Estimation and Statistical Inference

The post-onset sample was partitioned into distinct development regimes based on $\hat{\gamma}$. The OLS FE estimation was then executed within these data splits, allowing the slope coefficients $\theta_{k,L}$ and $\theta_{k,H}$ to vary flexibly based on whether a country-year observation fell above or below the established boundary.

Finally, we evaluated the statistical validity of the identified structural break. A cumulative effects test assessed the null hypothesis of regime equality (H_0: $\theta_{0,L} + \theta_{1,L} + \theta_{2,L} = \theta_{0,H} + \theta_{1,H} + \theta_{2,H}$) to determine if the estimated AI coefficients in the low and high regimes were statistically distinct as well as a test to see whether the cumulative effects in both regimes are non zero (H_0: $\theta_{0,L} + \theta_{1,L} + \theta_{2,L} = 0$ and H_0: $\theta_{0,H} + \theta_{1,H} + \theta_{2,H} = 0$). Moreover, to test the global existence of the threshold against a strictly linear null hypothesis, a Wild Cluster Bootstrap Sup-F test using 200 draws was performed. This bootstrap procedure explicitly respected the clustered nature of the panel errors, providing a conservative evaluation of the threshold's overarching statistical significance.

3.4.4 Robustness Checks

To ensure our primary OLS findings are not artifacts of residual dynamic panel bias, we employed the Anderson-Hsiao Instrumental Variable (AH-IV) estimator as a secondary robustness check for both RQ1 and RQ2 models.

Preliminary post-estimation diagnostics revealed that this specification exhibited significant second-order serial correlation in the differenced errors (AR(2) Woolridge test $p = 0.011$). This serial correlation renders the second lag endogenous. Therefore, to ensure strict causal validity, the deeper third lag of the dependent variable in levels, $y_{i,t-3}$, was used as the primary excluded instrument for the endogenous differenced lag $\Delta y_{i,t-1}$. This instrument satisfied the relevance restriction through its correlation with the endogenous regressor due to hysteresis in unemployment (Blanchard and Summers, 1986), and it also satisfied the exclusion restriction by remaining orthogonal to the future differenced error term.

Furthermore, the validity of this AH-IV estimator strictly required that the differenced error terms do not exhibit third-order serial correlation (AR(3)). Post-estimation diagnostic testing failed to reject the null hypothesis of no AR(3) in the differenced errors across the primary RQ1 specification ($p = 0.128$), the RQ1 robustness check ($p = 0.138$) and the RQ2 regime-split model ($p = 0.299$). From these results, we could confirm the validity of our excluded instruments as they remained strictly exogenous to the error term.

Because the inclusion of dynamic lags and Time Fixed Effects exhausted degrees of freedom, the First-Stage F-statistic for our threshold models attenuated to ≈ 9.60, falling below the Stock-Yogo critical value of of $F > 16.38$, we deployed the Anderson-Rubin (AR) test to guarantee valid inference regardless of instrument strength. The AR test remains robust and provides valid p-values even when the instruments are weak.

4. RESULTS AND DISCUSSION

4.1 Baseline effects of AI onset and Intensity

We estimate the specification for RQ1 to estimate whether AI onset and intensity affects the aggregate unemployment rate.

Table 6. Baseline Effects of AI Onset and Intensity

Variable	Coefficient (Standard Error)
d_S (AI patent activity onset time)	**0.033 (0.069)**
S × d_log_patents (AI_post_intensity)	**-0.0008 (0.09)**
L1_d_log_unemp (Lagged diff. log unemployment)	0.1869* (0.101)
d_inflation	2.000×10^{-4}*** (4.33×10^{-5})
d_Government Expenditure	0.0094** (0.003)
d_FDI	3.139×10^{-5} (3.83×10^{-5})
d_log_pop	0.993** (0.287)

(***) p<0.01, (**) p<0.05, (*) p<0.10

We see that neither the onset jump d_S_{it} nor the post onset AI development intensity term $AI\_post\_intensity_{it}$ have statistically significant coefficients. These results combine to suggest that contemporaneous early AI adoption and AI-related inventive activities does not generate detectable short-run aggregate labor effects within a linear framework and instead are consistent with a slow but gradual labour reallocation. Firms require time to convert technical possibility into deployable systems, reorganise workflows, and reallocate workers across tasks before innovation translates into observable employment outcomes.

However, the absence of significant linear effect does not preclude the presence of meaningful heterogeneity across countries. As established in Section 2, task-based frameworks predict that displacement and reinstatement forces operate with different magnitude depending on a country's structural maturity, sector composition and institutional capacity. Imposing a uniform slope coefficient across all countries will therefore obscure development-dependent regime differences. Furthermore, restricting the model to only contemporaneous effects ignores the fact that AI innovations often influence labour markets with a delay. These considerations motivate the threshold analysis in the next sections where we allow the effect of AI intensity to vary across development regimes and incorporate lag terms to capture delayed adjustment effects.

4.2 Threshold Identification and Regression

To assess whether the effect of AI intensity varies across development levels as measured by GDP per capita, we implement the threshold procedure outlined in section 3.3.2.

4.2.1. Threshold Identification

A preliminary OLS search identifies a threshold at approximately USD 24,564 which partitions the sample into low ($N = 159$; 26% of data) and high ($N = 453$; 74% of data) regimes.

However, the weak Hansen (1999) Sup-F test ($p = 0.865$) provides limited support for a nonlinear model specification. Several factors may explain this. The test is designed for static models with strictly exogenous regressors, whereas our specification includes a lagged dependent variable and distributed lags of AI intensity. Under these conditions, the test is likely to be underpowered, making it difficult to detect structural breaks even when they are present. Therefore, we interpret the global Sup-F result to be inconclusive instead of definitive. Further discussion of this limitation is explored in Section 5.

In light of this, we do not claim the estimated break represents a true structural discontinuity. Instead, we use the OLS-derived threshold as an exploratory partitioning divide to examine whether the relationship between AI and unemployment may differ across levels of development and all regime-based interpretations should be treated as exploratory and descriptive instead of evidence of structural nonlinearity. We show the existence of heterogeneity using a linear Wald test on the cumulative effects in the low and high regimes in the threshold regression model in the following section. Therefore, the OLS-based threshold is used as an indicative data-driven partitioning device, with emphasis on the magnitude and cumulative impact of AI intensity within each regime.

4.2.2 Threshold Regression Results

Using the OLS FE estimate of a break at USD 24,564 we estimate the distributed-lag specification across the two development regimes across countries with positive AI patent activity.

Table 7. Regime-Specific Distributed Lag Effects

Variable	Coefficient (Standard Error)
d_log_patents_low (diff. AI Activity in Low Regime)	**-0.081** (0.033)**
d_log_patents_high (diff. AI Activity in High Regime)	**0.001 (0.012)**
L1_d_log_patents_low (Lag 1 diff. AI Activity in Low Regime)	**-0.078** (0.025)**
L1_d_log_patents_high (Lag 1 diff. AI activity in High Regime)	**-0.022** (0.008)**
L2_d_log_patents_low (Lag 2 diff. AI Activity in Low Regime)	**-0.019 (0.013)**
L2_d_log_patents_high (Lag 2 diff. AI activity in High Regime)	**-0.019 (0.013)**
L1_d_log_Unemployment (Lagged diff. Log Unemployment)	0.406*** (0.061)
d_Inflation	0.006 (0.008)
d_Government Expenditure	0.005 (0.004)
d_FDI	-1.550×10^{-5} (4.530×10^{-5})
d_log_Population	1.549*** (0.514)

(***) p<0.01, (**) p<0.05, (*) p<0.10

Firstly, we verify that there is heterogeneity in the data. A Wald test testing the null of zero cumulative effects for the high and low region as well as the null of no cumulative effects in the low and high regime individually with the following results:

Table 8. Effects Test

Null Hypothesis	**P-Value**
$\theta_{0,L} + \theta_{1,L} + \theta_{2,L} = \theta_{0,H} + \theta_{1,H} + \theta_{2,H}$	**0.038****
$\theta_{0,L} + \theta_{1,L} + \theta_{2,L} = 0$	**0.009*****
$\theta_{0,H} + \theta_{1,H} + \theta_{2,H} = 0$	**0.089***

(***) p<0.01, (**) p<0.05, (*) p<0.10

We see that the Wald test shows there exists statistically significant heterogeneity in the lower and higher regimes, rejecting the null of equality across regimes of the joint cumulative effects. This justifies our approach to use the OLS threshold specification. An additional point of note is that the test rejects the null of no relationship between AI and unemployment in the lower development regime and the high development regime as well, however only at the 10% significance level for the latter.

Moving on to the coefficient estimates, Table 7 shows that in the lower development regime, the contemporaneous and lag 1 AI intensity terms are negative and significant, indicating that increases in AI patent intensity is associated with a reduction in unemployment both contemporaneously and with a 1-year delay. However we see that lag 2 AI intensity is insignificant. In the high development regime, the coefficient of $d\_log\_patents\_high$ is insignificant while the lag 1 term is negative and significant at the 5% level with a coefficient of -0.022 with an insignificant lag 2 term.

Combining the above results, we find that contemporary and lag 1 AI intensity are associated with a measurable reduction in unemployment in the low-development regime while the effects on more developed economies are weaker and are less precisely estimated. Additionally, the lag 2 terms are insignificant for both regimes.

4.3 Robustness Check

As a robustness check against potential Nickell Bias and simultaneity between Unemployment and AI, we present the re-estimation of results of the RQ1 and RQ2 regressions using the AH-IV approach, instrumenting unemployment with lag 3 unemployment with the following results:

Table 9. AH-IV Baseline Effects of AI Onset and Intensity

Variable	Coefficient (Standard Error)
L1_d_log_unemp (Lagged diff. log unemployment)	**1.034*** (0.098)**
d_S (AI patent activity onset time)	**0.085 (0.122)**
S × d_log_patents (AI_post_intensity)	**0.003 (0.119)**
d_inflation	2.000×10^{-4}*** (5.000×10^{-5})
d_Government Expenditure	0.005* (0.003)
d_FDI	3.000×10^{-5} (7.000×10^{-5})
d_log_pop	1.027*** (0.303)

(***) p<0.01, (**) p<0.05, (*) p<0.10

Table 10. AH-IV Regime-Specific Distributed Lag Effects

Variable	Coefficient (Standard Error)
d_log_patents_low (diff. AI Activity in Low Regime)	**-0.085** (0.040)**
d_log_patents_high (diff. AI Activity in High Regime)	**0.096 (0.013)**
L1_d_log_patents_low (Lag 1 diff. AI Activity in Low Regime)	**-0.064** (0.026)**
L1_d_log_patents_high (Lag 1 diff. AI activity in High Regime)	**-0.015 (0.012)**
L2_d_log_patents_low (Lag 2 diff. AI Activity in Low Regime)	**0.0082 (0.017)**
L2_d_log_patents_high (Lag 2 diff. AI activity in High Regime)	**-0.001 (0.0162)**
L1_d_log_Unemployment (Lagged diff. Log Unemployment)	0.967*** (0.096)
d_Inflation	0.016* (0.012)
d_Government Expenditure	0.002 (0.004)
d_FDI	-5.705×10^{-5} (7.179×10^{-5})
d_log_Population	1.284** (0.4265)

(***) p<0.01, (**) p<0.05, (*) p<0.10

We see that the IV estimates yield qualitatively similar results to the OLS specification, suggesting that any present endogeneity is marginal at most; however weak instrument concerns limit the ability to conclusively address potential bias.

With respect to RQ2, we examine potential dynamic panel bias using AH-IV estimation, yielding a break at a similar value of USD 25,783 which splits the sample into 180 low-regime observations

(29.4% of data) and 432 high-regime observations (70.6% of data). The similarity of the OLS (USD 24,564) and IV threshold further suggests that the split is not driven by endogeneity.

We see that the IV specification provides that in the low regime, the coefficients for the AI terms are negative and significant, similar to the OLS specification with mixed results in the high regime. We also see that the p-values and standard errors are wider as a consequence of using IV, and in particular a relatively weaker instrument in lag 3 unemployment.

One particular observation to note is that the coefficient for lagged unemployment is 0.18 in the OLS case and 0.967 in the IV case, suggesting that there may be attenuation bias in OLS estimates.

Overall, the robustness checks support the main conclusion that AI intensity has a stronger unemployment reducing effect in lower development countries, while effects in higher development economies remain limited.

4.4 Interpretation and Discussion

The results presented in the preceding sections can be interpreted in the framework outlined in section 2. In Acemoglu and Restrepo's model, automation generates both a displacement effect, which reduces labour demand in automated tasks and a reinstatement effect, where new tasks and productivity gains can restore employment, leading to a net ambiguous effect on unemployment and is therefore dependent on structural contexts.

In lower-development economies, the statistically significant negative cumulative effects suggest that labour-absorbing channels are more visible at the aggregate level. Because baseline automation exposure is lower in these economies (meaning there are fewer routine, highly paid manufacturing or clerical jobs to automate), the initial displacement margin is likely small. Interestingly, related work in Productivity J-curve (Brynjolfsson, Rock, and Syverson, 2021) suggests that the productivity effect of GPTs like AI typically requires a 5-to-10-year span to materialise at the macro level. Since our distributed lag model detects significant unemployment reductions within a compressed 2-year span, this leads us to believe that the short-term employment gains in these economies are consistent with reinstatement effects. Rather than just making existing processes faster, AI innovations in these settings likely accelerate the creation of entirely new digitally-enabled tasks and build new tech-deployment roles (such as prompt engineers, genAI scientists), rapidly absorbing displaced labour and creating demand for more labour in the short run.

Contrasting with high-development economies, where the marginally significant cumulative effect is consistent with offsetting forces. First, advanced economies possess mature service and manufacturing sectors with a high concentration of routine cognitive and administrative roles, making the initial displacement pressure from AI substantially larger. Secondly, the reinstatement effect in these economies may still create tasks, but the extent to which they create jobs to offset the displacement effect may be more limited. Firms in developed economies tend to have an already highly educated, digitally literate workforce. Therefore, the integration of AI may primarily drive internal task reallocation and intensive upskilling, where existing employees are expected to absorb and master new digitally-enabled tasks before an expansion of total headcount is considered. This aligns with empirical evidence from Acemoglu et al. (2022), who found that while AI adoption in US firms significantly shifts the composition of required skills, it does not generate a net increase in total firm-level employment.

Finally, the temporal distribution of these effects provides some suggestive evidence for a hyper-compressed “time-to-build” lag in modern AI adoption. The significance of the contemporaneous and one-year lagged terms in the lower-income regime and the one-year lagged term in the higher-income regime seems to signal a concentrated implementation shock that hits the labour market almost immediately. Interestingly, high-income countries only see a significant effect after a one-year lag, suggesting a slower restructuring process than developing economies. Consequently, the insignificance of the second lag suggests that the initial labour market restructuring phase happened and re-equilibrated within a compressed 12-to18-month window.

4.5 Policy Implications

The structural heterogeneity identified in our threshold model suggests that AI labour market policies may need to be tailored to a country’s specific stage of economic development, though the correlation nature of our evidence calls for caution in drawing direct causality.

For lower-development economies, the statistically significant reduction in unemployment suggests that policymakers could prioritise facilitating the reinstatement effect. The focus could be on lowering barriers to entry via subsidies for digital infrastructure projects and grants for AI-as-a-Service adoption, enabling workers to utilise AI more efficiently, as well as encouraging the growth of AI-related job roles. The creation of new tasks from these investments can outweigh labour displacement.

In contrast, for advanced economies like Singapore, the absence of a clear cumulative effect suggests that displacement and reinstatement forces may be in an offsetting equilibrium. Because our results indicate that these economies rely on an already tech-savvy workforce to absorb new tasks without increasing total headcount, the primary risk is structural mismatch of skills. Therefore, the upskilling-first approach of initiatives like Singapore's upcoming National AI Impact Program (Ministry of Digital Development and Information, 2026) could help to minimise the friction of internal task reallocation and prevent structural unemployment by ensuring that workers whose routine tasks become automated are equipped with the skills to move on to high-value, AI-enabled tasks in the same firm.

5. FUTURE RESEARCH AND LIMITATIONS

While this study establishes an empirical foundation for examining AI's labour market effects through both aggregate and threshold-dependent frameworks, several data-driven and methodological constraints suggest potential directions for future research to explore.

5.1 Model Selection and the Efficiency-Consistency Trade-off

A central challenge for this study is the trade-off between the precision of OLS FE and the consistency of AH-IV. Our AH-IV robustness check revealed a significantly higher coefficient for lagged unemployment of 0.967 compared to the OLS estimate of 0.18, suggesting that Nickell Bias may attenuate the true persistence of unemployment in OLS models.

However, our AH-IV robust models encountered weak instrument concerns, with first-stage F-statistics of 8.6 and 9.6 respectively, which are below the Stock-Yogo critical value of 16.38. While our AR weak-IV robust tests confirmed significance ($p = 0.001$ and $p = 0.0004$), standard asymptotic inference remains unreliable. This weakness stems from sparse AI patenting near adoption onsets, with 29% of the 612 post-onset observations having zero AI activity despite prior adoption.

Therefore, for a more consistent model, future work should pursue alternative instruments exploiting exogenous AI development shocks, employ jackknife IV methods (Chao et al., 2023), or leverage longer panels with post-2020 data.

5.2 Investigating Local versus Global Threshold Effects

The threshold analysis reveals an important interpretational nuance: while the regime-specific estimates suggest different cumulative AI effects between low- and high-income regimes, the Wild Cluster Bootstrap Sup-F test fails to reject the global linear null ($p = 0.865$). This suggests that the threshold effect may be sample-specific and not yet a globally robust feature. However, this result should be interpreted alongside the inherent data constraints of AI innovation. Our "lower-income" regime primarily consists of middle income and emerging economies, as true low-income nations lack AI patenting data to be included in this study.

The lack of a global structural break may therefore reflect the relatively narrow developmental gap between countries in our sample. Nonetheless, our findings remain highly relevant, as the sample is inclusive of a diverse range of both OECD and non-OECD high- and middle-income nations. Future research should therefore treat the estimated threshold as a sample-specific regime split supported by causal estimates, noting that a global structural break may only become statistically detectable as AI adoption data matures in the world's low-income economies.

5.3 Expanding Beyond 2020: The AI Boom Information Gap

The panel spans 1991-2020, capturing the early and some of the mid-stages of global AI adoption. However, since 2020, AI adoption has significantly intensified with the explosion of Generative AI (GenAI), potentially altering labour market dynamics beyond those observed in our historical sample.

Therefore, future research would be valuable to investigate:

- Do GenAI adoption produce sharper effects than the null Onset Jump effect ($p = 0.49$) found in this study?
- Does recent GenAI development exhibit similar development-dependent effects, or do lower deployment barriers eliminate regime differences?
- Does the hyper-compressed adjustment window identified in this study shorten even further with GenAI?

5.4 Investigating Alternative Measures of AI Exposure

Our study primarily uses AI patents as a measure of AI exposure. While patents capture codified inventive activity, they may not fully reflect actual adoption intensity, organisational integration, or worker exposure across sectors and countries. As a result, the estimated relationship may be sensitive to how AI exposure is operationalised, especially in cross country settings where patenting behaviour, legal systems, and innovation capacity differ substantially.

Future research should therefore combine patent based indicators with alternative measures such as task based exposure indices, firm level AI adoption surveys, and vacancy text measures that capture AI related skill demand. Using multiple proxies would improve construct validity, reduce measurement dependence on any single indicator, and allow stronger robustness checks on whether the estimated unemployment effects remain consistent across different definitions of AI exposure.

6. CONCLUSION

This paper investigates the impact of AI innovation on aggregate unemployment and examines how these effects differ across levels of economic development. Using a dynamic 36-country panel dataset (1991-2020), we investigate a development-based threshold model with a baseline OLS FE specification and an AH-IV specification for robustness check.

The baseline linear specification finds no statistically significant contemporaneous AI onset or post-onset AI patent intensity on unemployment, which suggests AI diffusion does not immediately lead to observable effects at the aggregate level.

Using a threshold model on heterogeneity yields richer results, with an identified structural break at approximately USD 24,564 GDP per capita. In lower income economies, AI innovation is correlated with a cumulative reduction in aggregate unemployment at the 1% significant level, driven by a concentrated implementation shock across contemporaneous and one-year lagged terms. Conversely, the cumulative effect in higher income economies is smaller and only marginally significant at the 10% level, appearing only after a one-year lag. The difference in cumulative effects is confirmed to be economically significant at the 5% level with a Wald test, providing strong evidence that AI's contrasting effects on labour markets differ with a country's economic development stage.

Furthermore, the concentration of these effects within a 12-to18-month window provides some support for a shorter "time-to-build" lag, signalling AI's departure from historical physical GPTs that require long-term investments in infrastructure for effects to materialise.

Several limitations remain, suggesting avenues for future research. Future research can extend this study by incorporating the post-2020 GenAI boom to verify whether our findings hold up against newer AI developments. Furthermore, while our regime-split is causally significant, the Sup-F test's failure to reject a global null indicates that the threshold effects may be sample-specific. Nonetheless, our results remain a robust representation of the divergence between emerging middle-income and advanced economies. Future research should expand this panel as AI diffusion reaches lower-income brackets to test for a broader global structural break.

Overall, the evidence suggests that AI's labour market effects are dynamic and dependent on development within the observed sample rather than uniformly disruptive globally. Regardless, this study still presents some compelling evidence of divergence to guide policymakers from countries in our sample on how to deal with AI's labour market effects. Policy interventions in developing economies could prioritise accelerating AI diffusion via targeted subsidies to maximise reinstatement effects, while advanced economies like Singapore require a shift to more aggressive upskilling initiatives to prevent structural unemployment.

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